



MARSHALL MANURUNG

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Undergraduate Informatics student with a strong focus on software development and a growing interest in artificial intelligence and data-driven problem solving. Experienced in designing and developing web and mobile applications, as well as applying machine learning techniques in academic projects. Demonstrates strong leadership, communication, and teamwork skills through teaching assistant roles, organizational involvement, and competitions. Highly adaptable and committed to building efficient, scalable, and intelligent technology solutions.

Education Level

Del Institute Of Technology - Toba, North Sumatera, Indonesia

Aug 2023 - Sep 2027 (Expected)

Bachelor, 3.27/4.00

- Awarded Best Male Student (University Level) – Even Semester, Academic Year 2024/2025
- Awarded Best Male Student, Pniel Dormitory – Academic Year 2023/2024
- Awarded Best Male Student, Louis Pasteur Dormitory – Academic Year 2024/2025

Projects

LPPM (Institute for Research and Community Service) - Del Institute of Technology

Sep 2025 - Dec 2025

System Analyst & Developer

- Conducted system requirement analysis and designed database structure for research and community service management
- Developed a web-based information system to streamline proposal submission and data management
- Built application features to support workflows, data processing, and reporting
- Contributed to end-to-end system development from planning to implementation

GPdI Sibulele–Balige Church - Balige, North Sumatra, Indonesia

Jan 2026 - Present

System Analyst & Developer

- Analyzed system requirements and designed database architecture for organizational data management
- Developed integrated web and mobile applications to support data management and improve accessibility
- Built features to manage and organize institutional data efficiently across platforms
- Contributed to end-to-end system development from design to implementation

Auction System - Del Institute of Technology

Nov 2025 - Dec 2025

Developer

- Developed a web-based auction platform enabling users to list items, place bids, and manage transactions
- Designed system workflow for bidding processes and user interactions
- Implemented features for real-time bidding and transaction management
- Focused on building an efficient and user-friendly system

Laundry Management Mobile Application - Del Institute of Technology

Mar 2026

Developer

- Independently developed an admin-side laundry management application, handling both front-end and back-end development
- Designed and implemented features for order management and transaction tracking
- Built system workflows to improve operational efficiency and data organization
- Developed a responsive and user-friendly interface to support daily administrative activities

Academic Project (Machine Learning Course)

Apr 2026 - May 2026

Computer Vision Developer (Team Project)

- Developed a computer vision model for classifying rice leaf diseases using image datasets relevant to agricultural conditions in North Sumatra
- Implemented Convolutional Neural Networks (CNN) for feature extraction and classification tasks
- Performed data preprocessing, including image resizing, normalization, and augmentation
- Collaborated in a team-based machine learning project, contributing to model development and evaluation

Work Experinece

Del Institute Of Technology - Toba, North Sumatera, Indonesia

Sep 2025 - Dec 2025

Teaching Assistant for Computer Networks

- Assisted in delivering networking laboratory sessions and mentored 50+ students in configuring routers and switches
- Guided students in understanding IP addressing and network fundamentals
- Supported troubleshooting during laboratory sessions

Del Institute Of Technology - Toba, North Sumatera, Indonesia

Feb 2026 - Present

Teaching Assistant – Data Structures and Algorithms

- Assisted in teaching data structures and algorithms and mentored 50+ students in programming and problem solving
- Supported coding labs and debugging sessions to improve students' implementation skills
- Guided students in analyzing algorithm efficiency and applying structured problem-solving techniques

Del Institute Of Technology - Toba, North Sumatera, Indonesia

Nov 2025 - Present

Treasurer – Informatics Student Association (HIMASTI)

- Managed budgeting, financial records, and reporting for organizational activities
- Ensured accountability and proper allocation of funds

Del Institute Of Technology - Toba, North Sumatera, Indonesia

Aug 2024 - May 2025

Student Leadership

- Mentored first-year students in academic adaptation and personal development
- Provided guidance on discipline, responsibility, and campus life

Del Institute Of Technology - Toba, North Sumatera, Indonesia

Oct 2025 - Nov 2025

Head of Event Division – HIMASTI Election Committee

- Led event planning and execution by coordinating committees and managing event flow
- Ensured smooth execution and adherence to schedule

Skills, Achievements & Other Experience

- **Competition** (2025): First Place Winner in HIMASTI Programming Competition 2025 at Del Institute of Technology, demonstrating strong algorithmic problem-solving and coding skills
- **Competition** (2025): Developed TrushCare, a smart waste classification system using computer vision and machine learning approaches in GEMASTIK Smart City Division
- **Competition** (2026): Third Place Winner in HIMASTI Datathon Competition at Del Institute of Technology, applying data analysis and machine learning techniques to solve real-world problems
- **ONMIPA** (2025): Selected as a university representative in ONMIPA Biology Regional Competition at Del Institute of Technology, demonstrating academic excellence and analytical skills
- **Projects** (2025): Wumpus World Intelligent Agent is an AI-based system that uses propositional logic and probabilistic reasoning to navigate a grid environment and make decisions under uncertainty.
- **Project** (2025): Local Search Optimization Project (N-Queens Problem) is a program developed to solve optimization problems using algorithms such as Hill-Climbing, Random-Restart, and Simulated Annealing, focusing on finding optimal solutions and analyzing performance.
- **Technical Skills:** JavaScript, TypeScript, Python, PHP, Java, Kotlin | React.js, Node.js, Laravel | MySQL, PostgreSQL | PyTorch, scikit-learn, NumPy, Pandas | Git, RESTful APIs | Mobile Development (Android)
- **Languages:** Indonesian (Native) | English (Intermediate)
- **All Certificates** 📄: A collection of certificates obtained from academic achievements, competitions, and organizational involvement.